

Parallel Computing

Programming with MPI – Part 4

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Objectives

To learn

- How to specify a list of processors in a file
- How to run in parallel two programs

- The machines to be used can be specified in a file:
`mpirun -np N --hostfile hosts code.exe`
where each line in the file `hosts` has the form of a machine name or IP address and the maximum number of processes that can be executed on that particular machine
- The current machine (on which `mpirun` has been launched) must be in the list contained in `hosts` as
`localhost`
Duplicate addresses give rise to an error
- Example of `hosts` file:

```
localhost slots=4
148.60.52.221 slots=4
148.60.41.222 slots=4
148.60.41.223 slots=4
```

The first 4 out of the N processes will be allocated to the first machine, the next 4 to the second machine, and so on

Access authorization (in `$HOME/.ssh/`)

- **Important:** It must be possible to connect to the remote machines without having to type in the password, using a public cryptographic key `id_rsa.pub` located in the `.ssh` directory. Create the key as follows

```
cd $HOME/.ssh  
ssh-keygen -t rsa
```

Choose default options by hitting the return key when requested

- In rooms 927 and 120, it is necessary to log in for the first time on one of the remote machines and type in your password

The address of the remote machine is displayed with the command `ifconfig`

In room 927: `148.60.52.nnn` where `nnn` are the last 3 digits of the computer name

In room 120: `148.60.41.n` where `n` is the last digit(s) of the computer name

Access authorization (in `$HOME/.ssh/`)

- If all machines share the same file system, add the key to the `authorized_keys` file in the `.ssh` directory

```
cat id_rsa.pub >> authorized_keys
```
- If the machines do not share the same file system, the modification to `authorized_keys` must be performed locally on each machine
- When running in parallel, the key contained in the `id_rsa.pub` is compared with the keys in `authorized_keys` and if it appears in the list, access is granted
- It is thus no longer necessary to enter the password when logging on the remote machine

Running two codes in parallel

- In MPI, it is possible to split the program into two sub-programs that will run in parallel under a single execution instruction
- Example : a main code, commonly called manager, will run on one processor, while the other, commonly called worker, will run on a large number N of processors. A simple execution instruction is

```
mpirun -np 1 manager.exe : -np N worker.exe
```

Tutorial: Manager-worker

- Write a `manager` code that first requests from the user the number N of random numbers between 0 and 1 to send, then sends a number to any other process that asks for it.
- Write a `worker` code that asks the manager to send a random number and prints it to the screen.
- When the N numbers have been sent, the `manager` sends a negative integer to all the other processes.
- The workers prints the number of positive numbers they have received and their sum.
- Run the manager and worker codes on
 - 1 the same host,
 - 2 two different hosts.
- Analyze the distribution of numbers in both cases.